

Skills, Inequality, and Referrals*

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1 Introduction

Referrals make up an integral part of the hiring process. Worker and firm surveys find that approximately half of all jobs are found through referrals (Topa, 2011). Their prevalence suggests referrals are an important channel for connecting workers with firms, transmitting valuable information to form good matches. Evidence shows that referrals have a higher probability of getting hired (Brown, Setren, & Topa, 2016), earn higher wages (Bayer, Ross, & Topa, 2008; Dustmann, Glitz, Schönberg, & Brücker, 2016; Hellerstein, Kutzbach, & Neumark, 2014, 2019; Schmutte, 2015), and are less likely to quit their jobs than non-referred hires (Dustmann et al., 2016; Friebel, Heinz, Hoffman, & Zubanov, 2023; Hellerstein et al., 2019).

The precise mechanism behind why job-seekers and firms alike resort to referrals remains elusive. One stream of literature originating from Montgomery (1991) suggests referrals help solve hiring firms' asymmetric information problem. Firms facing candidates of unknown productivity reduce uncertainty around the quality of a prospective

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hire via referrals, as highly productive workers bring in highly productive candidates. Granted the assumption of network homophily, a well-documented empirical consistency where individuals form ties more often with others who are similar to themselves across observable characteristics (McPherson, Smith-Lovin, & Brashears, 2006; McPherson, Smith-Lovin, & Cook, 2001), holds between highly productive individuals. Montgomery model conceptualizes how referrals convey information about productivity conditional on endogenous network formation. Supporting experimental evidence find productivity of referrals being higher than those of non-referred workers (Beaman & Magruder, 2012; Pallais & Sands, 2016). This result strengthens when referrals are made under performance contingent incentives as opposed to fixed bonuses which reduces the risk of social transfers (i.e., nepotism or favoritism) between referrer and referee (Beaman, Keleher, & Magruder, 2018; Beaman & Magruder, 2012). Past experiments looked at the referral-referrer productivity relationship keeping the network formation endogenous, with potential social transfer channels between the referrer and referrals (Beaman et al., 2018; Beaman & Magruder, 2012; Pallais & Sands, 2016; Witte, 2021). These make it difficult to isolate the characteristics of the information channel in referrals.

It remains an open empirical question whether some workers are connected more often with other productive workers, or given the same network, some referrers are better able to assess the productivity of their network members. We explore a new setup where the networks are fixed for all referrers, and social transfers are not of concern. What constitutes a referral in terms of the intensity of communication and reputational costs as a consequence of social transfers can be thought on a spectrum (Topa, 2019).¹ We consider a very simplified setup where a referral only consists of a private nomination of a network member whose is believed to be highly productive in a given task. There are no referrer-employer interactions and no possibility of social transfers. Once we abstract away from these elements, the referral exercise arguably resembles to one

¹On one extreme, letting someone know about a job opening requires no communication between the referrer and the employer with no reputational cost for the referrer. On the other extreme, recommending someone to an employer involves communication between the referrer and the employer, and comes at a large reputational cost for the referrer if the match does not work out.

of measuring productivity beliefs through nominated candidates. Understanding how referral decisions are made when the referral choice set is restricted and fully observable to researchers, without reputational costs and social transfers, is our main contribution.

While referral hiring may help increase job-candidate quality through homophily, it also propagates inequality: Employment and wages among friends correlate (Calvo-Armengol & Jackson, 2004; Calvo-Armengol & Jackson, 2007) and disadvantage qualified yet underrepresented worker groups (Beaman et al., 2018; Bolte, Immorlica, & Jackson, 2020; Jackson, 2021). Individuals are tied more frequently with those of comparable socioeconomic status (Chetty et al., 2022b), and differences in network composition lead to persistent differences in labor market outcomes, even when controlling for various productivity measures (Stansbury & Rodriguez, 2024). Our experiment takes place in a highly unequal society with significant class inequalities and limited class mobility (Angulo, Gaviria, Páez, & Azevedo, 2012; García, Rodríguez, Sánchez, & Bedoya, 2015), making salient the differences between social classes. Limited interaction across social classes due to spatial segregation is shown to drive at least some of the differences in outcomes (Chetty et al., 2022a), but we consider a setting where networks across social classes overlap. Anticipating differences in referral outcomes for low-SES even when spatial constraints are removed and productivity is accounted for, we introduce quota-like incentives as a treatment arm to increase referrals of low-SES. We design the treatment incentives in inspiration from the success of gender quotas in the affirmative action literature (Balafoutas & Sutter, 2012; Bertrand, Black, Jensen, & Lleras-Muney, 2019; Niederle, Segal, & Vesterlund, 2013).

A growing body of work emphasizes positive labor market returns for “noncognitive” or “soft” skills like patience, self-control, conscientiousness, teamwork, and critical thinking (Heckman & Kautz, 2012; Heckman, Stixrud, & Urzua, 2006; Lindqvist & Vestman, 2011; Weinberger, 2014). Deming (2017, 2023) separates these into interpersonal and intrapersonal skills, finding increasing returns to interpersonal (social) skills which are conceptually distinct but complementary to cognitive skill. Based on advances in this literature, we examine how well individuals can identify job-market relevant skills

in referrals. Past work show referrers can identify cognitive skill, but not a noncognitive skill like perseverance in a sample of low-educated day-workers in India (Beaman & Magruder, 2012). Evidence on social skill identification is limited. Yet, hiring firms report difficulties in observing job applicants’ social skills among their most important concerns, and applicants are willing to pay substantial sums to convey social skill feedback to employers (Bassi & Nansamba, 2022). We contribute to this literature by extending productivity screening to a highly educated sample for cognitive and social skills.

We separately measure cognitive and social skills from a diverse student body of peers who interact during semester-long university classes. We find that under performance-contingent incentives, referrers can screen cognitive skill but not social skill, providing another explanation for the large demand for communicating social skill to firms (Bassi & Nansamba, 2022). By fixing network formation and closing social transfer channels, we show that referrals reveal information about worker productivity as suggested by Beaman et al. (2018); Beaman and Magruder (2012); Pallais and Sands (2016), but only for skills than can be screened by referrers. We document a novel pattern in referral strategies when clean productivity signals are absent: Referrals often rely on a proxy linked to academic performance rather than the skill, highlighting limitations in precise skill screening. We find that low-SES individuals underperform in both skill measures, consistent with Falk, Kosse, Pinger, Schildberg-Hörisch, and Deckers (2021). However, controlling for productivity and network composition, we find limited bias against low-SES. Quota-like incentives improve outcomes for low-SES in the small proportion of cases where the bias exists.

Section 2 begins with the background and setting in Colombia. In Section 3 we present the design of the field experiment, including the skill assessment, referral and guessing tasks. In Section 4 we describe the data and procedures. Section 5 discusses the results of the experiment. Section 6 concludes. The Appendix presents and additional tables and figures as well as the experiment instructions.

2 Background and Setting

Inequality and Social Class. Colombia has consistently ranked as one of the most unequal countries in Latin America ([World Bank, 2024](#)), with the richest decile earning 50 times more than the poorest decile ([United Nations, 2023](#)). This economic disparity is reflected by a highly stratified society, where social classes remain separated by education, family background, lifestyle, occupation, power, and geographic residence.

Stratum system. In 1994, Colombia introduced a nationwide classification system dividing the population into 6 strata based on housing characteristics and neighborhood amenities. This system, initially designed for utility subsidies from higher strata (5 and 6) to support lower strata (1 to 3), now extends to university fees and social program eligibility. While imperfect -because it disregards the income of residents, this system largely aligns with and potentially reinforces existing social class divisions ([Guevara S & Shields, 2019](#); [Uribe-Mallarino, 2008](#)). We use this exogenous cutoff as the measure of social class in our experiment: Students in strata 1 to 3 are categorized as low-SES, and those in strata 4 to 6 as high-SES.

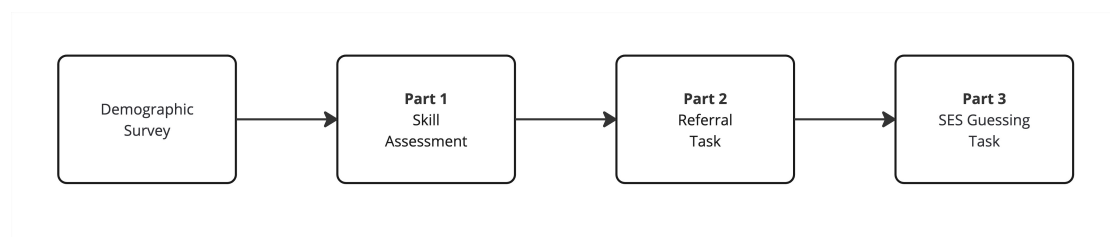
Universidad Autónoma de Bucaramanga. Our study takes place at UNAB, a medium-sized private university in Bucaramanga, Colombia, with approximately 10,000 enrolled students. UNAB’s student body is notably diverse, with slightly more than half of the students classified as low-SES (see Appendix Figure [A.1](#) for a detailed stratum distribution). This diversity reflects the broader trend in Colombian society, where lower middle-class families (stratum 2 and 3) prioritize university education for their children despite the financial burden ([Hudson & Library of Congress, 2010](#), p. 103). The university environment provides a unique opportunity for collaborative inter-class contact on equal status, whose positive effects on reducing discrimination are casually documented ([Lowe, 2021](#); [Mousa, 2020](#); [Rao, 2019](#)). We draw our sample from a cohort of students enrolled in two semester-long courses. Each course meets weekly for three-hour sessions. Attendance is mandatory, but enforcement varies across instructors. Both courses are university-wide graduation requirements which result in large variations in academic

programs and socioeconomic backgrounds across the classrooms.

3 Design

We designed a referral experiment between students from different social classes, and experimentally varied the incentives to refer a low-SES peer. The study design consists of a single experiment with three sequential parts with sessions organized at the classroom level (see Figure 1). The instructions are provided in Appendix B.

Figure 1: Experiment Timeline



Note: Participants first complete incentivized skill tests, then refer classmates for skills. In the final part, they guess the social class of their peers. This order is implemented in all sessions.

3.1 Skill Assessment

To understand the basis for referral decisions, we collect objective measures of cognitive and social skills. These two distinct skills are crucial for the labor market and suitable in a context with university students.² This approach ensures that we observe the distribution of skills in our sample, ask for referrals on these skills, and compare how well the same individuals refer peers for a specific skill. Participants perform two incentivized skill tests. They have 5 minutes to complete each test. We provide test-specific instructions and an example item before participants begin. Correctly solved items increase chances to earn a fixed bonus.³

²For the relationship between wages and social skills conditional on cognitive skills, see [Deming \(2017\)](#); [Edin, Fredriksson, Nybom, and Öckert \(2022\)](#); [Kuhn and Weinberger \(2005\)](#); [Weinberger \(2014\)](#).

³The tests are presented in a randomized order. No performance feedback is provided. Participants see one item at a time and cannot return to previous screens once they start a test. They are not required

Cognitive Skills. We use Raven’s Progressive Matrices to measure cognitive skills (Raven, 1936; Raven, Raven, & Court, 1976). Raven’s test is a well-established measure of fluid intelligence, i.e., an individual’s capacity to reason and solve problems in novel situations independent of past knowledge (Schilbach, Schofield, & Mullainathan, 2016). In this test, participants see series of images where there is a pattern with a piece that has been intentionally removed. They are tasked with choosing the piece that completes the pattern among available options. For each image, there is only one correct answer. We implement an 18-item version featuring increasingly difficult questions, with 6 response options for the first 9 items and 8 thereafter.

Social Skills. We measure social skills with the Multiracial Reading the Mind in the Eyes Test (MRMET) from Kim et al. (2022).⁴ The test is an established measure for the ability to recognize emotions in others, and it has been previously used in economic experiments (van Leeuwen et al., 2018; Weidmann & Deming, 2021). MRMET consists of photos of human faces portraying different emotions, cropped so that only the eye region is visible. Participants must choose the emotion that best describes the photo from the available answers. For each photo, there is only one correct answer and 4 response options. We administer the first 36 items in MRMET.

3.2 Referral Task

We create a referral task to measure bias against low-SES peers. For each skill, participants make incentivized referrals by nominating classmates. We first explain the measured skill accompanied by an example test item. On the following page, we provide an alphabetically ordered list of all classmates. Participants make three referral choices per skill. They are instructed to exclude themselves from referrals. A classmate may be nominated once per triad. The order in which participants refer for a skill test is to answer items and can skip them if they choose to do so. We elicit beliefs about performance after each test.

⁴We choose MRMET because it is a race- and gender-inclusive test suitable for application in non-WEIRD (Western, Educated, Industrial, Rich, Democratic) populations like the one we sample from. The test is based on the original RMET (Baron-Cohen, Wheelwright, Hill, Raste, & Plumb, 2001).

randomized. We incentivize referrals with classroom-level performance rankings. The three highest-scoring classmates are designated as the top 3 for a skill. Referrers are eligible for a fixed bonus for referrals among the top 3.⁵

Treatment. We have one between subject treatment, and it varies the top 3 selection. In the **Baseline** treatment, the top 3 selection is based solely on performance ranking, regardless of other participant characteristics. The **Quota** treatment modifies the top 3 selection to prioritize low-SES individuals. We reserve the first spot in the top 3 for the highest-scoring low-SES classmate, and assign the remaining two places based on performance. This guarantees at least one low-SES participant in the top 3 per skill. Participants are informed about the top 3 selection mechanism before making referral choices (Appendix Figure A.5 provides illustrations explaining the treatments). Assignment to the treatment is at the individual level within each classroom. This allows comparing the effect of the treatment while keeping the referral choice set constant.

3.3 SES Guessing Task

Participants make guesses about the anticipated SES of their classmates. We inform participants that a computer algorithm randomly selects three students belonging to strata 1, 2, or 3. They are tasked with nominating the people they believe the computer could choose at random (Appendix Figure A.6 provides the illustration explaining the task). Participants select three classmates from an alphabetically ordered list containing all their classmates. This task measures the ability to distinguish-SES independent of test performance, as SES identification is relevant to our study.

4 Sample, Incentives, and Procedure

Sample. We invited 849 UNAB undergraduate students to participate in the experiment. Our final sample consists of 713 individuals who completed the study protocol,

⁵We solve ties among the top 3 randomly. We describe only the top 3 selection mechanism and provide no feedback about the top 3 composition to participants.

resulting in an 83% participation rate. Table 1 presents key demographic characteristics and academic performance indicators across treatments (Appendix Table A.1 compares the sample and missing students). The sample is well-balanced between the Baseline and Quota conditions. We observe no statistically significant differences in any of the reported variables (all p values > 0.1), indicating balanced randomization. The sample is characterized by a majority of low-SES students. About one-third of the sample are first-generation college students. The gender distribution is balanced. The mean GPA of 3.95 is consistent across both treatments.

Incentives. Participants could earn bonuses worth 100,000 Pesos (about 26 US Dollars) in each part of the experiment. In the first part, we incentivized performance in the skill tests. 20% of participants were eligible for the bonus. We randomly picked one skill test for each eligible participant and drew a number between 1 and 100. The participant received the bonus if the percentage of correct answers in the selected test exceeded the drawn number. Chances of earning the bonus increased with each correctly solved question by 5.5% ($=1/18$) for the Cognitive Skill test and by 2.78% ($=1/36$) for the Social Skill test. In the second part, we incentivized referrals among the top 3 performers. 40% of participants were eligible for the bonus. We randomly selected one skill test and one referral for each eligible participant. The participant received the bonus if their referral was among the top 3. In the third part, we incentivized the correct identification of low-SES classmates. 20% of participants in each classroom were eligible for the bonus. We randomly selected one guess for each eligible participant. The participant received the bonus if their guess correctly identified a low-SES classmate.

Procedure. Data collection occurred during the last weeks of April 2024. Our local partner, the Social Bee Lab, coordinated scheduled classroom visits and recruited research assistants to administer the experiment. Students present in class on the scheduled visit dates participated. Each classroom visit constituted a separate session. There were in total 35 sessions. Participants accessed the Qualtrics-based experiment using their smartphones during these visits. The median time to complete the survey was 20 minutes, with a compensation of \$26 for 117 lottery winners.

Table 1: Comparison of demographic characteristics and academic outcomes across treatment conditions

	Baseline	Quota	<i>p</i>
Low-SES	59%	55%	0.297
Cognitive skill (out of 18)	10.04	10.27	0.322
Social skill (out of 36)	18.45	18.50	0.886
GPA	3.95	3.95	0.828
Entry Exam	61.85	62.17	0.638
Age	19.33	19.02	0.228
Female	52%	47%	0.195
First Generation	34%	37%	0.386
Ethnic Minority	1%	3%	0.133
Rural Community	30%	27%	0.308
Has Scholarship	1%	1%	0.916
# Semesters at UNAB	3.18	3.17	0.916

Note: Low-SES, female, first generation, ethnic minority, rural, and has scholarship represent binary outcomes and *p*-values for these variables are from two-sample tests of proportions. For continuous variables (age, GPA, cognitive skill, social skill, entry exam, and semester), *p*-values are from two-sample t-tests with equal variances. All reported *p*-values are two-tailed. Sample sizes vary slightly across measures due to missing data.

5 Empirical Analysis

Can peers screen cognitive and social skills?

Our first goal is understanding whether higher skilled individuals get more referrals. Because every referrer nominates 3 classmates per skill, analyzing only the extensive margin, i.e., whether an individual gets a referral, is not very informative.⁶ We consider the percentage share of referrals from individuals in Baseline condition as our dependent variable. This approach combines the intensive and extensive margins and also makes comparisons across classrooms with different sizes easier.⁷ Formally, we define the percentage share of referrals received by individual i from participants j in classroom c and in Baseline condition ($\forall j \in B_c$) for skill $s \in \{Cognitive, Social\}$ as:

$$y_{ic}^s = \frac{\sum_{j \neq i} r_{ijc}^s}{n_c - \mathbb{1}(i \in B_c)} \times 100 \quad (1)$$

where n_c represents the number of participants in the Baseline condition in classroom c . The indicator r_{ijc}^s takes value 1 if participant j in the Baseline condition refers individual i for skill s , and 0 otherwise, and require both i and j to be in the same classroom c . The denominator $n_c - \mathbb{1}(i \in B_c)$ accounts for the maximum possible referrals that individual i could receive. If i is in the Baseline condition ($\mathbb{1}(i \in B_c) = 1$), we subtract one from n_c to account for the self-referral restriction.⁸ This normalized measure represents the percentage of potential referrals actually received by each individual, adjusting for classroom size and treatment status. By construction, $y_i^s \in [0, 100]$ for all

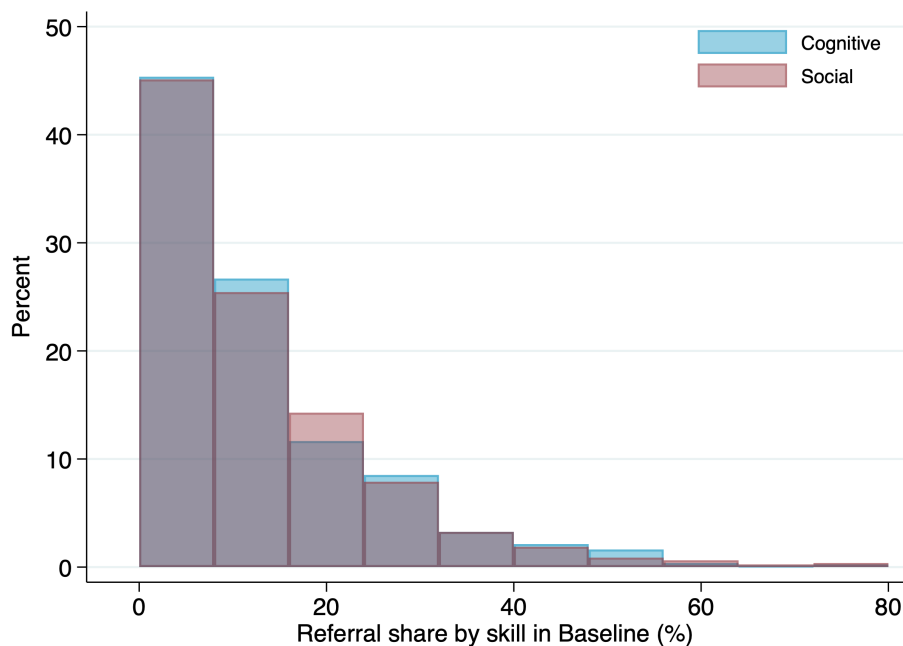
⁶Only 86 of the 849 students (10%) never get a referral for either skill.

⁷The number of participants in a classroom mechanically drives the number of total referrals that could be received by an individual. By normalizing referrals we focus on differences within classrooms.

⁸33.8 percent of participants in the sample for cognitive and social skills self-referred, while explicitly instructed not to do so. In Appendix Table A.4 compares the outcomes of those who self-refer. Self-referrers are more likely to be low-SES, and have significantly lower cognitive skill (0.2 SD) and GPA (0.25 SD). We rule out the hypothesis that self-referrers nominate themselves strategically. As self-referrers are not informative and add noise to our estimates, we drop these instances from our paired referral-referrer sample in subsequent analyses. Self-referring participants' remaining referral choices are kept in the dataset.

c , and we can compare referrals across classrooms of different sizes. Figure 2 presents the distribution of our dependent variable.

Figure 2: Referrals received at Baseline



Note: The figure shows the percentage share of referrals received across cognitive and social skills from participants in Baseline condition. It is calculated as the number of actual referrals received divided by the maximum possible referrals a student could have received.

Under performance pay in the Baseline condition, higher performing classmates in skill tests should collect more referrals if classmates can successfully screen skills. Our independent variables are the standardized skill test scores. We estimate referral percentage shares y_i^s :

$$y_i^s = \beta_0^s + \beta_1^s Skill_i^s + \epsilon_i^s \quad (2)$$

Table 2 illustrates our first findings. The comparison of interest is the point estimates for different skills. In column (1), a one standard deviation increase in cognitive skill causes a 1.2 percentage point increase in the share of referrals received. On a base rate

of 13%, this is a modest increase of 9 percent. In column (2), 95% confidence intervals rule out that a one standard deviation increase in the social skill result in more than a 1 percentage point difference in the share of referrals received. Participants have difficulties screening skills in the Baseline condition, with modest recognition of cognitive and no recognition at all of social skills.

Table 2: Share of referrals received conditional on skill

	(1)	(2)
	Cognitive	Social
Skill	1.197** (0.479)	0.037 (0.510)
Constant	12.986*** (0.653)	13.049*** (0.521)
Observations	665	665

Note: Classroom-level clustered standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Dependent variables are the percentage of referrals received relative to all referrals. “Skill” refers to standardized test scores for cognitive and social skills. Sample restricted 665 individuals for whom we have complete administrative and experimental data.

In the absence of a clean skill-signal from their peers, participants may have referred classmates using proxies of skills. We define proxies as peer beliefs about strong correlates for skills. A potential proxy for cognitive skill (i.e., “smart students”) would be the “students with good grades” in the classroom, as measured by GPA. The idea that grades signal cognitive skill is a common belief among researchers and practitioners alike. Yet, cognitive skill and grades are far from perfectly correlated (Heckman & Kautz, 2012; Heckman et al., 2006), and screening with such beliefs may not lead to good referrals. Indeed, GPA correlates very weakly with skills in our sample (see Appendix Table A.2). We capture the screening behavior using proxies by including the standardized GPA

scores of referrals as an independent variable. We reestimate referral percentage shares for the Baseline condition:

$$y_i^s = \beta_0^s + \beta_1^s Skill_i^s + \beta_2^s GPA_i^s + \epsilon_i^s \quad (3)$$

Table 3 illustrates our findings. The comparison of interest is the difference between point estimates for skills and GPA. In column (1), a one standard deviation increase in cognitive skill causes a 0.9 percentage point increase in the share of referrals received when controlling for GPA. On a base rate of 12.8%, this is a comparable increase in magnitude of about 7 percent to our previous estimate in Table 2, and suggests cognitive skills have an independent effect on referrals. However, a one standard deviation increase in GPA causes a substantial 3.9 percentage point increase in the share of referrals received when controlling for cognitive skill. This is an increase of more than four times in terms of magnitude (30 percent) when compared to cognitive skill, and suggestive of the extent to which academic performance is easier to screen among peers. In column (2), 95% confidence intervals rule out that a one standard deviation increase in the social skill result in more than a 1 percentage point difference in the share of referrals received. This is consistent with our previous estimate confirming participants cannot screen social skills. On the other hand, a one standard deviation increase in GPA causes a substantial 3.4 percentage point increase in the share of referrals received when controlling for social skill. This is a slightly smaller increase of 26 percent in the share of referrals for social skill. For both skills, we find strong evidence that grades act as a proxy for referral decisions. In the next section we expand on the diversity in referral choices to differentiate between referrers using GPA proxy and others.

Table 3: Share of referrals received conditional on skill and academic performance

	(1) Cognitive	(2) Social
Skill	0.873* (0.467)	-0.278 (0.460)
GPA	3.949*** (0.664)	3.429*** (0.581)
Constant	12.806*** (0.662)	12.891*** (0.636)
Observations	665	665

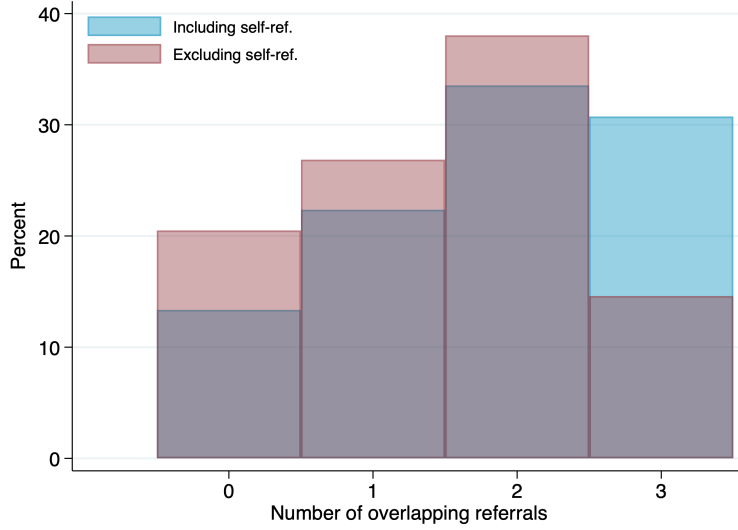
Note: Classroom-level clustered standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Dependent variables are the percentage of referrals received relative to all referrals. “Skill” refers to standardized test scores for cognitive and social skills. GPA is standardized to mean zero and unit variance.

Referral strategies

Diversity in referral choices can vary substantially as participants refer 3 individuals per skill, and can nominate the same individual twice. In Figure 3, we show the share of referrers disaggregated by the number of times a they nominated the same classmate for both skills. Referral choices are highly concentrated around the same individuals: Only 20 percent of participants pick all unique classmates for their 6 referral choices, and most participants make two common referrals. Excluding self-referrals underestimates the share of participants who have 3 identical referral choices, and the diversity in referral choices gets even lower with self-referrals ⁹

⁹In Appendix Table A.5 we compare the characteristics of referrers make unique referrals to those who made at least one common referral. Results suggest minimal differences in GPA, skills, and social class.

Figure 3: Share of common referrals



Note: The figure shows the distribution of overlapping referrals with and without self-referrals. The first bar (value of 0) indicates the share of participants with 6 unique referrals. The last bar (value of 3) indicates the share of participants with 3 identical referral choices.

With such a large share overlapping referral choices, it is possible that participants believed classmates with a higher score in one skill would also have a higher score in the other. Would such beliefs be accurate? There is only modest ($\rho = 0.267$) correlation between the two skills (see Appendix Table A.2). To understand whether making overlapping referrals is strategic, we turn to the incentives. Participants were incentivized to pick the top 3 performers in their classroom for each skill to earn a fixed bonus. Looking at the characteristics of top skilled participants in Appendix Table A.6, we find that conditional on being among the top 3 for one of the skills, only 1 in 3 participants were in the top 3 for both skills. This suggests making more than 1 common referral across skills would decrease the chances to win the bonus.

A competing explanation for the amount of overlapping referral choices coupled with the notable difficulties in screening skills would be that individuals who refer classmates

twice for both skills are worse at observing peer skills. This implies the underlying heterogeneity in skill identification results in differential referral strategies where participants with a good signal for a skill choose to refer classmates only once for that skill, and those without a good signal use the grade proxy and refer classmates for both skills. We can test both hypotheses in our data: If “twice” referrers -defined as those who refer an individual for both skills- are better at screening at least one of the skills, this gives more credence to beliefs about correlated skills. On the other hand, if “twice” referrers are worse in skill identification compared to single-referrers and use GPA proxy for referrals, we can infer that they have no additional information about skills.

We compare the outcomes of participants who receive referrals twice from their classmates to those who receive referrals only once per skill. Formally, let indicator r_{ijc}^{twice} take value 1 if individual j referred individual i for both skills. The percentage share of referrals received by individual i from participants in classroom c and in Baseline condition ($\forall j \in B_c$) is:

$$y_{ic}^{twice} = \frac{\sum_{j \neq i} r_{ijc}^{twice}}{n_c - \mathbb{1}(i \in B_c)} \times 100 \quad (4)$$

where n_c represents the number of participants in the Baseline condition in classroom c . The indicator r_{ijc}^{twice} takes value 0 if participant j in the Baseline condition does not refer individual i for both skills. The denominator $n_c - \mathbb{1}(i \in B_c)$ accounts for the maximum possible “twice” referrals that individual i could receive as before. Similarly, let $r_{ijc}^{s,single}$ take value 1 if individual j referred individual i only for skill s . The percentage share of “single” referrals received by individual i from participants in classroom c and in Baseline condition ($\forall j \in B_c$) for skill s is:

$$y_{ic}^{s,single} = \frac{\sum_{j \neq i} r_{ijc}^{s,single}}{n_c - \mathbb{1}(i \in B_c)} \times 100 \quad (5)$$

and it follows that for any s , percentage share of “single” and “twice” referrals received by individual i from participants in classroom c and in Baseline condition ($\forall j \in B_c$)

must add up to the total share of referrals received:

$$y_{ic}^s = y_{ic}^{s, single} + y_{ic}^{twice} \quad (6)$$

we regress Equation 3 for our three new dependent variables and report our findings in Table 4. The comparison of interest is the skill and GPA estimates across columns. In column (1), we find that a one standard deviation increase in GPA causes a 3.2 percentage point increase in the share of “twice” referrals received when controlling for skills. This is a staggering 43 percent increase on a base rate of 7.4%. 95% confidence intervals rule out that a one standard deviation increase in either skill result in more than a 1 percentage point difference in the share of “twice” referrals received. Participants who nominate the same individuals for both skills make referrals based on academic performance and cannot observe classmates with higher cognitive or social skills. For participants who receive only cognitive skill referrals, in column (2), we find that a one standard deviation increase in GPA causes a 0.8 percentage point increase in the share of referrals when controlling for cognitive skill. A one standard deviation increase in cognitive skill causes a comparable 1 percentage point increase in referrals when controlling for GPA. These are respectively 15 and 19 percent increases in the share of referrals received, and suggest participants are able to screen higher skilled peers when referring only for cognitive skill. The lower base rate of 5.4% compared to 7.4% in column (1) suggest less than half of referrals came from “single” referrals. GPA estimate is three times smaller in magnitude compared to column (1), and suggests a smaller weight put in the grades proxy. Nevertheless, the comparable magnitudes of GPA and cognitive skill point estimates still suggest participants refer peers with higher grades much more often than the correlation between the two supported by the data ($\rho = 0.085$). There is heterogeneity in skill identification even for those who refer a peer once for cognitive skill. For participants who receive social skill referrals only, in column (3), 95% confidence intervals rule out that a one standard deviation increase in social skill or GPA result in more than a 1 percentage point difference in the share referrals received. These further support our previous finding that peers cannot screen

social skills in our sample, and attempts to screen social skills by ceasing the use of the GPA proxy seems to fail. Taken together, there are three distinct referral strategies we find in our sample. A majority of individuals cannot screen skills, refer instead using the GPA proxy, and nominate the same individuals for both skills. Those who refer an individual only for cognitive skill can identify this skill, and drive the entirety of the results in terms of peer skill recognition. Still, they confound the cognitive skill signal with academic performance, and put similar weights on the two. Finally, those who refer an individual only for social skill can neither screen social skill or use the GPA proxy. In the subsequent section, we analyze referrals from the perspective of social class while accounting for the referral strategies described in this part.

Table 4: Share of “twice” versus “single” referrals received conditional on skills and academic performance

	(1)	(2)	(3)
	Twice	Cognitive Only	Social Only
GPA	3.172*** (0.464)	0.801** (0.391)	0.260 (0.334)
Cognitive Skill	-0.042 (0.416)	1.006*** (0.270)	
Social Skill	-0.353 (0.304)		0.086 (0.381)
Constant	7.407*** (0.524)	5.400*** (0.386)	5.485*** (0.376)
Observations	665	665	665

Note: Classroom-level clustered standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The dependent variable in column (1) is the percentage share of “twice” referrals received from the same referrer, in column (2) “single” referral share for cognitive skill, and in column (3) for social skill. Independent variables are the respective standardized test scores for skills and GPA.

Social class bias across referral strategies

Based on the referral strategies found in the previous section, we document the existence of a social class bias in referrals when controlling for skills and academic performance at Baseline. Our dependent variables are the percentage shares of referrals received at Baseline as defined in Equation 6, and we include a social class dummy for the participant receiving the referrals. We estimate for our three dependent variables:

$$y_i^s = \beta_0^s + \beta_1^s GPA_i + \beta_2^s Skill_i^s + \beta_3^s SES_i + \epsilon_i^s \quad (7)$$

Table 5 summarizes our findings. The comparison of interest is the SES estimates for the three referral strategies. In column (1), controlling for skills and GPA, the point estimate for low-SES is not statistically significant. Skill and GPA estimates are robust to the inclusion of this variable and remain close to those in Table 4. For participants who receive cognitive skill referrals only in column (2), we find that being low-SES causes a 2 percentage points decrease in the share of referrals when controlling for cognitive skill and GPA. This is a substantial 31 percent difference in the share of referrals received, confirming participants are biased against low-SES peers when referring only for cognitive skill. Skill and GPA estimates are robust to the inclusion of this variable, but the decrease in the cognitive skill point estimate from 1.006 in Table 4 to 0.869 with the inclusion of SES dummy suggest the two are confounders. This is supported by the data, where low-SES underperform in cognitive skill (see Appendix Figure A.3a). GPA and low-SES are not confounders and there are no differences across social classes in terms of GPA (see Appendix Figure A.2). For participants who receive social skill referrals only, in column (3), the point estimate for low-SES is not statistically significant. GPA estimate remains as before. The decrease in the social skill point estimate with the inclusion of SES dummy suggests the two are confounders, and this is also supported by the data: Low-SES underperform also in social skill (see Appendix Figure A.3b). Taken together, we document a low-SES bias for cognitive skill referrals only. In the next section we explore whether the social class bias survives when we consider referrals together, and the effects of the Quota treatment on low-SES referrals.

Table 5: Share of “twice” versus “single” referrals received conditional on skills, academic performance, and social class

	(1)	(2)	(3)
	Twice	Cognitive Only	Social Only
GPA	3.170 ^{***}	0.797 ^{**}	0.260
	(0.462)	(0.386)	(0.334)
Cognitive Skill	0.000	0.869 ^{***}	
	(0.411)	(0.261)	
Social Skill	-0.306		0.047
	(0.315)		(0.372)
Low-SES	0.799	-2.017 ^{***}	-0.549
	(0.939)	(0.711)	(0.610)
Constant	6.948 ^{***}	6.558 ^{***}	5.800 ^{***}
	(0.742)	(0.656)	(0.558)
Observations	665	665	665

Note: Classroom-level clustered standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The dependent variable in column (1) is the percentage share of “twice” referrals received from the same referrer, in column (2) “single” referral share for cognitive skill, and in column (3) for social skill. Independent variables are the respective standardized test scores for skills and GPA.

Social class bias and the Quota treatment

In the following empirical specification, we document whether there is a social class bias when referrals are grouped, and whether the Quota treatment causes referral shares of low-SES participants to change when controlling for skills and academic performance. We hypothesize that the Quota treatment should increase referrals to low-SES classmates because of the additional incentive to refer low-SES. The dependent variable is the percentage share of referrals received as defined for the Baseline treatment in Equation

1, now extended to the referrals from the Quota treatment. It is trivial to see y_{ic}^s can also be calculated for the Quota treatment as participants in every classroom are randomized into either treatment. Now, every participant is observed twice in the data, and we add a treatment dummy to indicate whether the referrals came from participants in the Baseline or the Quota treatment. We also add a social class dummy for the participant receiving the referrals to our specification. We estimate:

$$y_i^s = \beta_0^s + \beta_1^s Quota_i + \beta_2^s SES_i + \beta_3^s (Quota_i \times SES_i) + \beta_4^s Skill_i^s + \beta_5^s GPA_i + \epsilon_i^s \quad (8)$$

Table 6 illustrates our findings. Our comparison of interest is the effect of the Quota treatment on low-SES peers. In column (1) for cognitive skill, we find that being low-SES decreases the share of referrals recieved by about 1.2 percentage points when controlling for the skill and academic performance in the Baseline condition. This goes down to about 1.5 percentage points in the Quota condition. These differences are not statistically significant, but their sign suggests a relatively large bias against low-SES classmates: A one standard deviation increase in cognitive skill has less than half this magnitude. This finding is coherent with the bias against low-SES classmates found in the previous section, but suggests it is not large enough to not carry over to all cognitive skill referrals considered together. In column (2) for social skill, we find that being low-SES has no statistically significant effect on the share of referrals recieved when controlling for the skill and academic performance. The sign for the point estimate suggest low-SES get slightly more referrals in the Baseline condition, and the Quota reduces this share. The lack of statistical significance in the SES main effect and interaction term suggests considerable noise in both estimates.

Table 6: Share of referrals received by referrer treatment, controlling for skill, academic performance, and social class

	(1)	(2)
	Cognitive	Social
Quota	-0.073 (0.755)	0.299 (0.716)
Low-SES	-1.230 (1.079)	0.364 (1.282)
Quota $\times$ Low-SES	-0.167 (1.117)	-0.835 (1.181)
Skill	0.594 (0.448)	0.201 (0.426)
GPA	3.184*** (0.517)	2.819*** (0.493)
Constant	13.551*** (0.986)	12.706*** (0.952)
Observations	1,330	1,330

Note: Classroom-level clustered standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The dependent variables are the percentage share of referrals received for cognitive and social skills. Quota is a dummy for the referrals received from classmates in the Quota treatment. Low-SES is a dummy for participant's socioeconomic status. Remaining independent variables are the respective standardized test scores for skills and GPA.

Effects of the Quota treatment across referral strategies

Effects of the social class bias gets diluted across different referral strategies. A large proportion of participants - “twice” referrers- who struggle with skill identification and screen for skills using the academic performance proxy. But there are no SES differences

for GPA in our sample, and past evidence suggests low-SES may even outperform high-SES at the university level in some samples [CITE MANU WORK](#). When referrals are made with academic performance in mind, it seems reasonable not to observe a negative bias against low-SES. Then what about skills, knowing that high-SES score higher in both measures? We observe a bias in undersampling from equally well performing low-SES only for “single” cognitive skill referrals, where referrers have a cleaner signal about the skill compared to “single” social skill referrals. We expect the Quota treatment be effective in increasing referrals for low-SES only in a scenario where the skill can be screened, and turn toward our classification of different referral strategies to test this hypothesis. To get clearer estimates for the effects of the Quota treatment on low-SES referrals, we re-estimate the shares of “twice” and “single” referrals. Following the same logic in the section before, we observe every participant twice in each specification, and add a treatment dummy to indicate whether the referrals came from referrers in the Baseline or the Quota treatment. We keep the social class dummy and regress Equation 8 for the three dependent variables.

Table 7 illustrates our findings. The comparison of interest is the SES of the participant receiving the referrals and the effect of the Quota treatment across “twice” and “single” referral types. In column (1), for participants who refer the same peers twice using the academic performance proxy, we find no statistically significant effect of participant SES or the Quota on the referrals share when controlling for skills and academic performance. The signs on the point estimates suggest that directionally low-SES get more referrals in the Baseline treatment and that the Quota reduces this share. For participants who make only cognitive skill referrals, in column (2), we find that being low-SES in the Baseline treatment reduces the percentage share of referrals received by 2.1 percentage points when controlling for the skill and academic performance. This is a very large effect size which translates to a decrease in referral share by 32 percent on a base rate of 6.6%, and is identical to the one found in Table 5. In turn, the Quota treatment increases referrals to low-SES by 1.42 percentage points when controlling for the skill and academic performance. This is also a large effect size that results in an

increase in low-SES referral share by 22 percent. For participants who make only social skill referrals, in column (2), we find no statistically significant effect of participant SES or the Quota on the referrals share when controlling for the skill and academic performance. These are in accordance with our previous findings that social skills cannot be identified and provide no reason to warrant social class bias. Taken together, we find support for our hypothesis that there is a bias against low-SES peers for the skill that is well-identified by peers, and in which low-SES underperform. We find no evidence of a bias when referrals are made based on academic performance where both social classes perform equally well. Reassuringly, we find the bias in cognitive skill referrals is at least partially alleviated by the Quota treatment. Because there is remarkable heterogeneity in the ability to detect SES for both social classes (see Appendix Figure [A.4](#)), we find this increase in low-SES referrals satisfying for our sample.

Table 7: Share of “twice” and “single” referrals received by referrer treatment, controlling for skill, academic performance, and social class

	(1)	(2)	(3)
	Twice	Cognitive Only	Social Only
Quota	0.436 (0.817)	-0.509 (0.598)	-0.136 (0.523)
Low-SES	0.857 (0.920)	-2.074*** (0.722)	-0.510 (0.613)
Quota $\times$ Low-SES	-1.584 (1.159)	1.417** (0.656)	0.750 (0.717)
Cognitive Skill	-0.079 (0.374)	0.658*** (0.201)	
Social Skill	0.062 (0.283)		0.158 (0.312)
GPA	2.322*** (0.330)	0.858*** (0.312)	0.502* (0.278)
Constant	6.952*** (0.727)	6.591*** (0.654)	5.765*** (0.562)
Observations	1,330	1,330	1,330

Note: Standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The dependent variable in column (1) is the percentage share of “twice” referrals received from the same referrer, in column (2) “single” referral share for cognitive skill, and in column (3) for social skill. Quota is a dummy for the referrals received from classmates in the Quota treatment. Low-SES is a dummy for participant’s socioeconomic status. Remaining independent variables are the respective standardized test scores for skills and GPA.

6 Conclusion

We find the effects of referrals on skill screening in a diverse university setting with peer interaction. When referral choice sets are fixed and social transfer channels are closed, students in our sample successfully refer higher cognitive skilled peers. Building on past experiments where referrals contain information about productivity (Beaman et al., 2018; Beaman & Magruder, 2012; Pallais & Sands, 2016), our results present a nuanced picture in the efficiency of referrals for screening ability. We find that the skill signal is often confounded with the signal of a proxy, and the magnitude of this misidentification is very large. Because the academic performance proxy proved not accurate, referrals using the proxy resulted in poor referral skill performance overall. The nature of the productivity information conveyed via referrals is very much dependent on the underlying beliefs about the correlation between the proxy and the skill.

Most individuals referred directly using the proxy instead of focusing on the skills. For referrals to be efficient in screening for the skill, referrers may first need to believe they have a clean skill signal. Forcing referrals when skill signal is not clear may have resulted in the proxy strategy taking over the majority of referrals. Making a referral optional as in Beaman et al. (2018); Beaman and Magruder (2012) in one treatment arm could have helped testing this claim, and we leave this for future research. The fact that we observe the proxy strategy proves performance contingent incentives are the right way to elicit referrals in experimental settings.

The evidence for a social class bias is intertwined with the different referral strategies. Low-SES students get a smaller share of referrals only for “single” referrals for cognitive skill, and not for other types of referrals when controlling for productivity. The interpretation we offer is grounded in data and specific to our context. Referrals made with the academic proxy successfully identify high GPA peers. In our university setting, peers know who in their classroom are good students, and social classes are equally good in terms of GPA. This creates little room for bias against low-SES students. In contrast, low-SES students have significantly lower scores in both skill measures. The fact that

we observe a bias only in cognitive skill suggests underperforming low-SES students may be overrepresented, in line with stereotype conceptualization by [Bordalo, Coffman, Gennaioli, and Shleifer \(2016\)](#), but only in the skill which can be effectively screened for.

Our findings contribute to understanding referral mechanisms by showing that even in settings where networks are fixed and social transfers eliminated, referrals remain an imperfect tool for screening worker skills. The effectiveness of referrals depends critically on referrers' ability to observe and accurately assess specific skills, suggesting important implications for how firms structure their referral programs and highlighting the need for complementary screening mechanisms.

References

- Angulo, R., Gaviria, A., Páez, G. N., & Azevedo, J. P. (2012). Movilidad social en colombia. *Documentos CEDE*.
- Balafoutas, L., & Sutter, M. (2012). Affirmative action policies promote women and do not harm efficiency in the laboratory. *science*, *335*(6068), 579–582.
- Baron-Cohen, S., Wheelwright, S., Hill, J., Raste, Y., & Plumb, I. (2001). The “Reading the Mind in the Eyes” Test Revised Version: A Study with Normal Adults, and Adults with Asperger Syndrome or High-functioning Autism. *The Journal of Child Psychology and Psychiatry and Allied Disciplines*, *42*(2), 241–251. (Publisher: Cambridge University Press) doi: 10.1017/S0021963001006643
- Bassi, V., & Nansamba, A. (2022). Screening and signalling non-cognitive skills: experimental evidence from uganda. *The Economic Journal*, *132*(642), 471–511.
- Bayer, P., Ross, S. L., & Topa, G. (2008). Place of work and place of residence: Informal hiring networks and labor market outcomes. *Journal of political Economy*, *116*(6), 1150–1196.
- Beaman, L., Keleher, N., & Magruder, J. (2018). Do Job Networks Disadvantage Women? Evidence from a Recruitment Experiment in Malawi. *Journal of Labor Economics*, *36*(1), 121–157. doi: 10.1086/693869

- Beaman, L., & Magruder, J. (2012). Who Gets the Job Referral? Evidence from a Social Networks Experiment. *American Economic Review*, 102(7), 3574–3593. doi: 10.1257/aer.102.7.3574
- Bertrand, M., Black, S. E., Jensen, S., & Lleras-Muney, A. (2019). Breaking the glass ceiling? the effect of board quotas on female labour market outcomes in norway. *The Review of Economic Studies*, 86(1), 191–239.
- Bolte, L., Immorlica, N., & Jackson, M. O. (2020). The role of referrals in immobility, inequality, and inefficiency in labor markets. *arXiv preprint arXiv:2012.15753*.
- Bordalo, P., Coffman, K., Gennaioli, N., & Shleifer, A. (2016). Stereotypes. *The Quarterly Journal of Economics*, 131(4), 1753–1794.
- Brown, M., Setren, E., & Topa, G. (2016). Do informal referrals lead to better matches? evidence from a firm’s employee referral system. *Journal of Labor Economics*, 34(1), 161–209.
- Calvo-Armengol, A., & Jackson, M. O. (2004). The effects of social networks on employment and inequality. *American economic review*, 94(3), 426–454.
- Calvó-Armengol, A., & Jackson, M. O. (2007). Networks in labor markets: Wage and employment dynamics and inequality. *Journal of economic theory*, 132(1), 27–46.
- Chetty, R., Jackson, M. O., Kuchler, T., Stroebe, J., Hendren, N., Fluegge, R. B., ... others (2022a). Social capital II: determinants of economic connectedness. *Nature*, 608(7921), 122–134.
- Chetty, R., Jackson, M. O., Kuchler, T., Stroebe, J., Hendren, N., Fluegge, R. B., ... others (2022b). Social capital I: measurement and associations with economic mobility. *Nature*, 608(7921), 108–121.
- Deming, D. J. (2017). The growing importance of social skills in the labor market. *The quarterly journal of economics*, 132(4), 1593–1640.
- Deming, D. J. (2023). Multidimensional human capital and the wage structure. *National Bureau of Economic Research*.
- Dustmann, C., Glitz, A., Schönberg, U., & Brücker, H. (2016). Referral-based job search networks. *The Review of Economic Studies*, 83(2), 514–546.

- Edin, P.-A., Fredriksson, P., Nybom, M., & Öckert, B. (2022). The rising return to noncognitive skill. *American Economic Journal: Applied Economics*, 14(2), 78–100.
- Falk, A., Kosse, F., Pinger, P., Schildberg-Hörisch, H., & Deckers, T. (2021). Socioeconomic status and inequalities in children’s iq and economic preferences. *Journal of Political Economy*, 129(9), 2504–2545.
- Friebel, G., Heinz, M., Hoffman, M., & Zubanov, N. (2023). What do employee referral programs do? measuring the direct and overall effects of a management practice. *Journal of Political Economy*, 131(3), 633–686.
- García, S., Rodríguez, C., Sánchez, F., & Bedoya, J. G. (2015). La lotería de la cuna: La movilidad social a través de la educación en los municipios de colombia. *Documentos CEDE*.
- Guevara S, J. D., & Shields, R. (2019). Spatializing stratification: Bogotá. *Ardeth. A Magazine on the Power of the Project*(4), 223–236.
- Heckman, J. J., & Kautz, T. (2012). Hard evidence on soft skills. *Labour economics*, 19(4), 451–464.
- Heckman, J. J., Stixrud, J., & Urzua, S. (2006). The effects of cognitive and noncognitive abilities on labor market outcomes and social behavior. *Journal of Labor economics*, 24(3), 411–482.
- Hellerstein, J. K., Kutzbach, M. J., & Neumark, D. (2014). Do labor market networks have an important spatial dimension? *Journal of Urban Economics*, 79, 39–58.
- Hellerstein, J. K., Kutzbach, M. J., & Neumark, D. (2019). Labor market networks and recovery from mass layoffs: Evidence from the great recession period. *Journal of Urban Economics*, 113, 103192.
- Hudson, R. A., & Library of Congress (Eds.). (2010). *Colombia: a country study* (5th ed.). Washington, D.C: Federal Research Division, Library of Congress: For sale by the Supt. of Docs., U.S. G.P.O. Retrieved from the Library of Congress, <https://www.loc.gov/item/2010009203/>.
- Jackson, M. O. (2021). Inequality’s economic and social roots: the role of social networks

- and homophily. *Available at SSRN 3795626*.
- Kim, H., Kaduthodil, J., Strong, R. W., Germine, L., Cohan, S., & Wilmer, J. B. (2022). *Multiracial Reading the Mind in the Eyes Test (MRMET): an inclusive version of an influential measure* (preprint). doi: 10.31219/osf.io/y8djm
- Kuhn, P., & Weinberger, C. (2005). Leadership skills and wages. *Journal of Labor Economics*, 23(3), 395–436.
- Lindqvist, E., & Vestman, R. (2011). The labor market returns to cognitive and noncognitive ability: Evidence from the swedish enlistment. *American Economic Journal: Applied Economics*, 3(1), 101–128.
- Lowe, M. (2021). Types of contact: A field experiment on collaborative and adversarial caste integration. *American Economic Review*, 111(6), 1807–1844.
- McPherson, M., Smith-Lovin, L., & Brashears, M. E. (2006). Social isolation in america: Changes in core discussion networks over two decades. *American sociological review*, 71(3), 353–375.
- McPherson, M., Smith-Lovin, L., & Cook, J. M. (2001). Birds of a feather: Homophily in social networks. *Annual review of sociology*, 27(1), 415–444.
- Montgomery, J. D. (1991). Social Networks and Labor-Market Outcomes: Toward an Economic Analysis. *American Economic Review*.
- Mousa, S. (2020). Building social cohesion between christians and muslims through soccer in post-isis iraq. *Science*, 369(6505), 866–870.
- Niederle, M., Segal, C., & Vesterlund, L. (2013). How costly is diversity? affirmative action in light of gender differences in competitiveness. *Management Science*, 59(1), 1–16.
- Pallais, A., & Sands, E. G. (2016). Why the Referential Treatment? Evidence from Field Experiments on Referrals. *Journal of Political Economy*, 124(6), 1793–1828. doi: 10.1086/688850
- Rao, G. (2019). Familiarity does not breed contempt: Generosity, discrimination, and diversity in delhi schools. *American Economic Review*, 109(3), 774–809.
- Raven, J. C. (1936). *The Performances of Related Individuals in Tests Mainly Educative*

- and Mainly Reproductive Mental Tests Used in Genetic Studies* (PhD Thesis). University of London (King's College).
- Raven, J. C., Raven, J., & Court, J. H. (1976). *Standard progressive matrices: sets A, B, C, D & E*. Oxford, UK: Oxford Psychologists Press.
- Schilbach, F., Schofield, H., & Mullainathan, S. (2016). The Psychological Lives of the Poor. *American Economic Review*, 106(5), 435–440. doi: 10.1257/aer.p20161101
- Schmutte, I. M. (2015). Job referral networks and the determination of earnings in local labor markets. *Journal of Labor Economics*, 33(1), 1–32.
- Stansbury, A., & Rodriguez, K. (2024). The class gap in career progression: Evidence from US academia. *Working Paper*.
- Topa, G. (2011). Labor markets and referrals. In *Handbook of social economics* (Vol. 1, pp. 1193–1221). Elsevier.
- Topa, G. (2019). Social and spatial networks in labour markets. *Oxford Review of Economic Policy*, 35(4), 722–745.
- United Nations. (2023). *Social panorama of latin america and the caribbean 2023: labour inclusion as a key axis of inclusive social development*. ECLAC and United Nations. Retrieved from <https://www.cepal.org/es/publicaciones/68702-panorama-social-america-latina-caribe-2023-la-inclusion-laboral-como-eje-central>
- Uribe-Mallarino, C. (2008). Estratificación social en bogotá: de la política pública a la dinámica de la segregación social. *Universitas humanistica*(65), 139–172.
- van Leeuwen, B., Noussair, C. N., Offerman, T., Suetens, S., van Veelen, M., & van de Ven, J. (2018). Predictably Angry—Facial Cues Provide a Credible Signal of Destructive Behavior. *Management Science*, 64(7), 3352–3364. doi: 10.1287/mnsc.2017.2727
- Weidmann, B., & Deming, D. J. (2021). Team Players: How Social Skills Improve Team Performance. *Econometrica*, 89(6), 2637–2657. doi: 10.3982/ECTA18461
- Weinberger, C. J. (2014). The increasing complementarity between cognitive and social skills. *Review of Economics and Statistics*, 96(5), 849–861.

Witte, M. (2021). Why do workers make job referrals? experimental evidence from ethiopia. *Working Paper*.

World Bank. (2024). *Regional poverty and inequality update spring 2024* (Poverty and Equity Global Practice Brief). Washington, D.C.: World Bank Group. Retrieved from <http://documents.worldbank.org/curated/en/099070124163525013/P17951815642cf06e1aec4155e4d8868269>

A Additional Figures and Tables

A.1 Additional Figures

Figure A.1: Stratum distribution of the sample

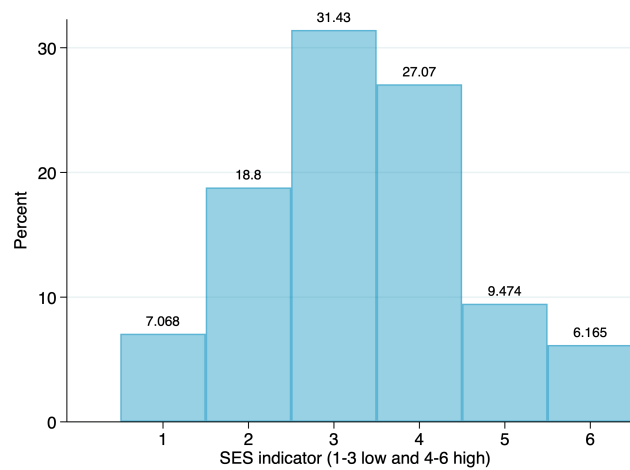
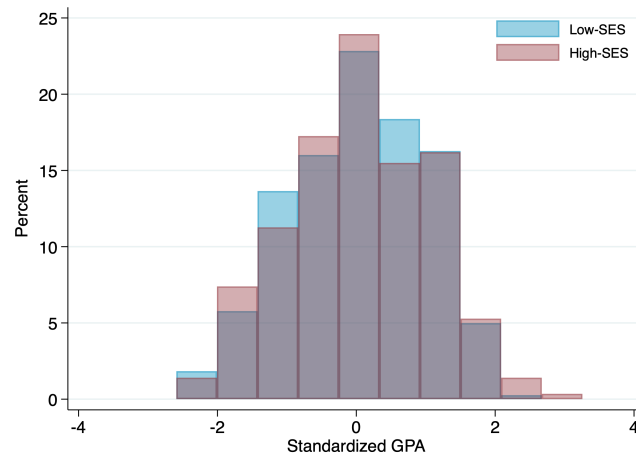
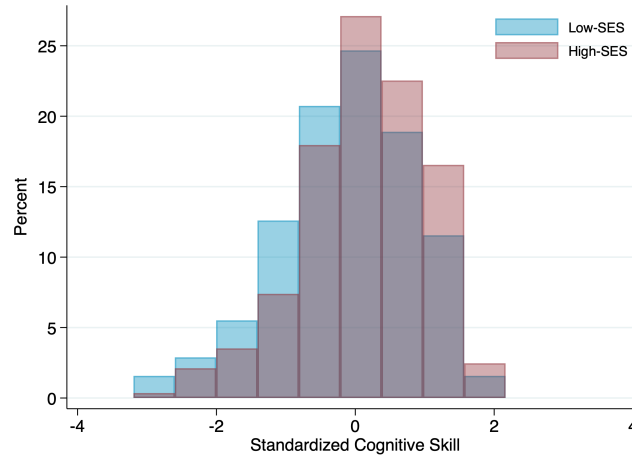


Figure A.2: GPA by SES

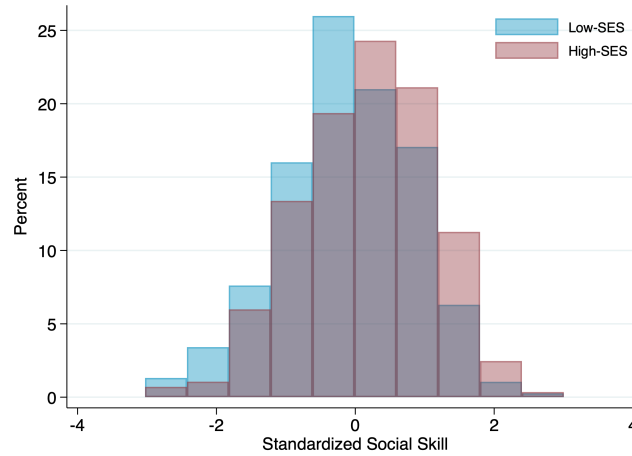


Note: Figure shows the distribution of GPA across SES. There are no significant differences in the mean standardized GPA scores between high-SES and low-SES participants (t test $p = 0.695$).

(a) Cognitive Skill by SES

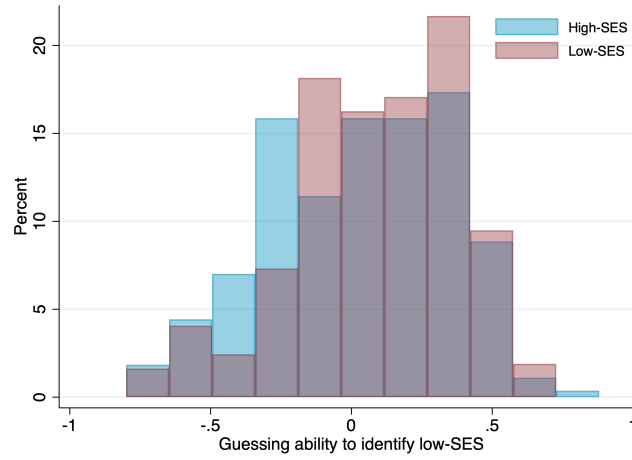


(b) Social Skill by SES



Note: Figures show the respective distribution of cognitive and social skills across SES. High-SES outperform Low-SES in both skills (t tests have p values < 0.001). We can visually verify that larger share of high-SES in quantiles above median for both skills.

Figure A.4: Distribution of guessing ability across SES



Note: Figure shows the distribution of the guessing ability across SES. We calculate the guessing ability as the share of succesful low-SES guesses minus the expected probability of randomly drawing low-SES in class c . A score of 0 indicates an accuracy as good as random draws, below 0 drawing worse than chance, and above 0 better than chance. There are significant differences in the mean guessing ability between high-SES ($M = 0.022$, $SD = 0.325$, $n = 271$) and low-SES participants ($M = 0.093$, $SD = 0.302$, $n = 369$), $t(638) = -2.85$, $p = 0.005$, $d = 0.226$. Low-SES participants have higher guessing ability compared to their high-SES counterparts, with a mean difference of 7 percentage points.

A.2 Additional Tables

Table A.1: Comparison of Variables Between the Sample and Missing Students

	Sample	Missing	p
Referral share (both skills)	0.127	0.043	0.000
GPA (standardized)	0.044	-0.273	0.001
Entry Exam (standardized)	0.028	-0.168	0.046
# Semesters at UNAB	3.171	3.188	0.884
Age	19.182	20.287	0.001
Female	49.8%	48.5%	0.788
Ethnic Minority	2.1%	4.4%	0.114
Rural Community	28.8%	31.6%	0.501
Has Scholarship	0.8%	0.7%	0.899

Note: Values for female, ethnic minority, rural community, and scholarship represent percentages. All other variables represent means. p -values for gender, ethnic, rural, and scholarship are from two-sample tests of proportions. For all other variables, p -values are from two-sample t-tests with equal variances. All tests compare the sample and missing students. All reported p -values are two-tailed.

Table A.2: Correlation between GPA and skills

	GPA	Cognitive	Social
GPA	1.000		
Cognitive	0.085	1.000	
Social	0.093*	0.267*	1.000

Note: Pairwise correlation between GPA and skills. * indicates significance at the 5% level with Bonferroni correction. Sample is restricted to 655 participants with complete administrative and experimental data.

Table A.3: Between-Classroom Variation in Academic Programs

Statistic	Most common program share
Mean	0.424
Standard Deviation	0.216
10th percentile	0.174
25th percentile	0.292
Median	0.345
75th percentile	0.533
90th percentile	0.696
# classrooms with share 1	3
Most diverse classroom	0.154
# classrooms	35

Note: Table shows the distribution of academic programs across classrooms, measured by the share of students from the most common program in each classroom. Three classrooms are completely homogeneous (share = 1). In the median classroom, the most common program accounts for 34.5% of students. The most diverse classroom has only 15.4% of students in the same program. Data based on 849 students across 35 classrooms.

Table A.4: Characteristics of self-referrers

	No self-referral	Any self-referral	Δ	p
GPA	0.132 (1.003)	-0.120 (0.966)	0.252	0.002
Cognitive Skills	0.087 (0.988)	-0.118 (1.023)	0.205	0.013
Social Skills	0.034 (1.003)	-0.038 (0.959)	0.072	0.374
Low-SES	0.605 (0.490)	0.511 (0.501)	0.094	0.021
N	440	225	665	
Share (%)	66.2	33.8	100	

Note: Table compares standardized scores between participants who self-referred at least once ($N = 225$) and those who did not ($N = 440$). Positive differences indicate higher scores for those who never self-referred. p -values from two-sided t -tests (GPA, Cognitive Skill, Social Skill) and proportion test (Low-SES). The results suggest self-referrers have significantly lower cognitive skills and GPA, and are more likely to be low-SES. Standard deviations in parentheses, samples restricted to participants with complete administrative and experimental data.

Table A.5: Characteristics of participants who make overlapping referrals

	Unique referrals	Overlapping referrals	Δ	p
GPA	0.057 (0.983)	0.045 (1.009)	0.012	0.903
Cognitive Skill	0.110 (1.005)	0.024 (0.979)	0.086	0.371
Social Skill	-0.014 (0.938)	0.033 (0.981)	-0.047	0.621
Low-SES	0.530 (0.501)	0.597 (0.491)	-0.067	0.164
N	132	512	644	
Share (%)	20.5	79.5	100	

Note: Table compares characteristics between participants who made at least one overlapping referral ($N = 512$) to those who did not ($N = 132$, 20.5%). Overlapping referrals indicate cases where a participant referred the same classmate once for cognitive or social skills. Positive differences indicate higher scores for those who made no overlapping referrals. The results suggest minimal differences across all variables. p -values from two-sided t-tests (GPA, Cognitive Skill, Social Skill) and proportion test (Low-SES). Standard deviations in parentheses, sample restricted to participants with complete administrative and experimental data.

Table A.6: Characteristics of Top Performers and Referrals

	Cognitive		Social		Both
	Top 3	Referrals	Top 3	Referrals	Top 3
Cognitive Skill	1.223 (0.419)	0.112 (1.009)	0.383 (0.922)	0.058 (1.015)	1.201 (0.458)
Social Skill	0.357 (0.923)	0.086 (0.996)	1.340 (0.395)	0.042 (1.009)	1.391 (0.453)
GPA	0.277 (0.990)	0.251 (1.021)	0.264 (1.046)	0.212 (1.004)	0.551 (0.897)
Low-SES	0.457 (0.500)	0.532 (0.499)	0.456 (0.500)	0.555 (0.497)	0.500 (0.507)
N	129	1,759	114	1,775	36
Share (%)	20.0	100	17.7	100	5.6

Note: Table shows characteristics of students ranked in the top 3 of their classroom and average characteristics of referred students, by skill. Standard deviations in parentheses. Sample restricted to participants with complete administrative and experimental data. All continuous variables are standardized.

B Experiment

We include the English version of the instructions used in Qualtrics. Participants saw the Spanish version. Horizontal lines indicate page breaks, and clarifying comments are inside brackets.

Please enter the password:

[classroom-specific password sent to each participant the day before data collection]

Welcome

Welcome to this study organized by the Social Bee Lab. You have been invited to participate in a survey where you can make a series of decisions. The study takes approximately 20 minutes to complete. During the study, you should not communicate with any other students. If you have any questions at any time, please raise your hand. One of the assistants will help you privately.

In this study, you can win bonus money depending on your choices. In total, we will draw [classroom-specific number equal to 40% of class size] bonuses of 100.000 pesos among the participants of this classroom. It is also possible for the same person to win more than one voucher. The following screens will detail how the bonus draw will be conducted. The UNAB finance office will make the payment of the vouchers through Nequi.

All your decisions in this survey will be anonymized. Therefore, the answers you provide will not affect your grades in this class or your records at the university. We will use your personal information to determine the bonus allocation, but after that, we will remove any data that identifies you.

This survey has several parts. Each of these parts has specific instructions. Please read the instructions for each part carefully because they describe how you can earn bonuses. This study has been approved by the [omitted for anonymous review] on the condition that all the information we provide is true and all the bonuses we offer are real. On the next screen, we present you with an informed consent form that you must accept to participate in this study.

Informed Consent

You have been invited to participate in a study to learn more about how people make decisions in common scenarios.

This study is conducted by [omitted for anonymous review] and the Social Bee Lab at UNAB. The purpose of this study is to broaden our understanding of how people make decisions.

Participation in this study is voluntary. You may opt-out at any time. No known risks are associated with your participation in this project beyond those of everyday life. Apart from the monetary bonuses that will be drawn, participation has no direct benefits.

The Social Bee Lab is in charge of data collection. Your answers in this study are anonymous and will not be shared with anyone. In addition to your answers, UNAB will provide the Social Bee Lab with administrative records of your courses and your university entrance exam score. Your records, decisions, and your identity will be kept strictly confidential. Data about you collected within the scope of the study are used for scientific purposes only and are treated as strictly confidential. The Social Bee Lab will anonymize your data, and the researcher will analyze it without knowing your identity. All data generated will be stored on the researcher's computer. You have the right to access your personal data and request its deletion. You can exercise this right by contacting the researcher.

If something is unclear or you have any questions, you can contact [omitted for anonymous review].

If you have questions about your rights as a participant, you can contact [omitted for anonymous review].

By continuing to the next screen, you agree to participate in this study.

Before you start, please answer these four questions.

What is your gender?

[Male, Female]

What is the socio-economic stratum to which your family belongs?

[Stratum 1 to Stratum 6]

What is your father's highest acquired level of education?

[Primary school, High school, Technical school, Undergraduate, Graduate, Postgraduate, Not applicable.]

What is your mother's highest acquired level of education?

[Primary school, High school, Technical school, Undergraduate, Graduate, Postgraduate, Not applicable.]

Part 1

You will now participate in two quizzes, each lasting five minutes. Please try to answer

them to the best of your ability.

We will allocate up to [classroom-specific number equal to 20% of class size] bonuses of 100.000 pesos in this first part. The steps to allocate the bonuses for Part 1 are explained below.

[classroom-specific illustrations explaining the incentive structure]

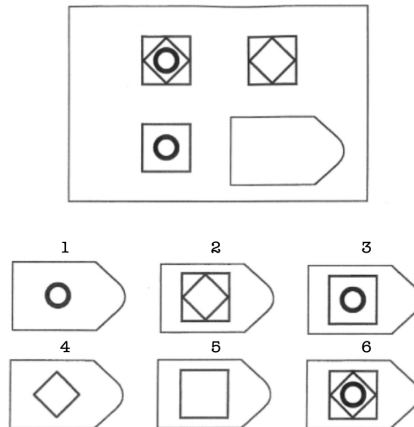
[random assignment to either cognitive or social skills test]

Test - Cognitive Skill

In this test, you will see a series of images. Below is an example of the images you will solve. At the top of each image, there is a pattern with a piece that has been removed. Your task is to choose which of the six pieces completes the pattern correctly. For each image, there is only one correct piece. Look at the following example:

First, notice a square in the upper left, the upper right, and the lower left. Also, notice that the circle is eliminated when one moves from the upper left to the upper right. Finally, the rhombus is eliminated when moving from the upper left to the lower left. Therefore, the correct piece should eliminate the circle and the rhombus, leaving only a square. So, the correct answer is piece 5.

To give your answer to each image, you must choose the correct option and then continue to the next screen. After giving your answer you cannot go back.



You will have 5 minutes to complete the test, which consists of 18 images to solve. The percentage of correct answers will determine your chances of winning one of the 100.000 pesos bonuses if you are chosen for the drawing.

Are you ready?

Your 5 minutes will start as soon as you move to the next screen.

Problem 1

[screenshot of Raven's matrix]

[After participants submit an answer, a new matrix appears on the screen. The sequence of matrices is the same for all participants. Participants cannot return to a

previous screen. Participants do not have to provide answers for all 18 matrices.]

You have finished the test. You can proceed to the next screen.

How did you do on the test?

If we randomly choose 10 participants from this classroom, how many people do you think solved fewer correct problems than you?

[Slider from 0 to 10]

Test - Emotions

In this test, you will see a series of photographs. Below is an example of the pictures you will see. In each picture, you will see the eyes of a person. Below the picture, you will see four possible emotions that this person is feeling. Your task is to choose which of the four emotions correctly describes what the person is feeling. For each picture, there is only one emotion. Look at the following example:



[Happy, Disappointed, Shocked, Worried]

In this case, the correct answer is: Shocked.

To give your answer to each picture, you must choose the correct option and then continue to the next screen. After giving your answer you will not be able to go back.

You will have 5 minutes to complete the test, which consists of 36 photographs to solve. The percentage of correct answers you get will determine your chances of winning one of the 100.000 pesos bonuses if you are chosen for the drawing.

Are you ready?

Your 5 minutes will start as soon as you move to the next screen.

Photograph 1: Choose the word that best describes the photograph

[photo from Multiracial Reading the Mind in the Eyes Test]

[After participants submit an answer, a new photo appears on the screen. The sequence of photos is the same for all participants. Participants cannot return to a previous screen. Participants do not have to provide answers for all 36 photos.]

You have finished the test. You can proceed to the next screen.

How did you do on the test?

If we randomly choose 10 participants from this classroom, how many people do you think solved fewer correct photographs than you?

[Slider from 0 to 10]

Part 2

At the beginning of this study, all participants took two tests, one on cognitive ability and one on emotions. In this part, we will ask you to recommend the people who in your opinion will score the best on each test.

You may recommend 3 people per test, but you may not recommend yourself.

We will allocate up to [classroom-specific number equal to 40% of class size] bonuses of 100.000 pesos for Part 2. The steps for allocating bonuses are explained below.

[random assignment to either quota or baseline condition]

[classroom-specific illustrations explaining the incentive structure depending on assign-

ment to either baseline or quota conditions]

Figure A.5: Illustrations for the two conditions



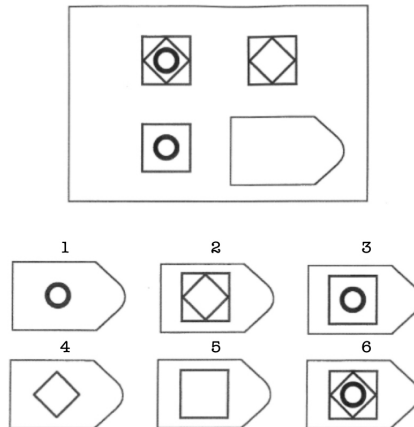
[random assignment to either cognitive or social skills referral task]

Recommendation - Cognitive Skill

All participants took a test to identify the missing pattern in each image, as in the example below. This test is used to measure general intelligence.

Next, we will present you with a list of the names of all the students in this room. We will ask you to recommend the three people you think will score the highest on the general intelligence test.

If you are chosen by the computer, each of your recommendations in the top 3 increases your chances of winning one of the 100.000 pesos bonuses.



Select the students in this classroom who you consider to have the highest scores on the general intelligence test. (Select 3 students)

[Classroom-specific list of all classmate names visible on one screen. Participants have to pick 3 classmates to continue. Picking their own name invalidates their choices.]

Recommendation - Emotions

All participants took a test where they had to identify the emotion that best described the expression of each image as in the example below. This test is used to measure social skills.



Next, we will present you with a list of the names of all the students in this room. We will ask you to recommend 3 people you think will score the highest on the social skills test.

If you are chosen by the computer, each of your recommendations in the top 3 increases your chances of winning one of the 100.000 pesos bonuses.

Select the students in this classroom who you consider to have the highest scores on the social skills test. (Select 3 students)

[Classroom-specific list of all the names visible on one screen. Participants have to pick 3 classmates to continue. Picking their own name invalidates their choices.]

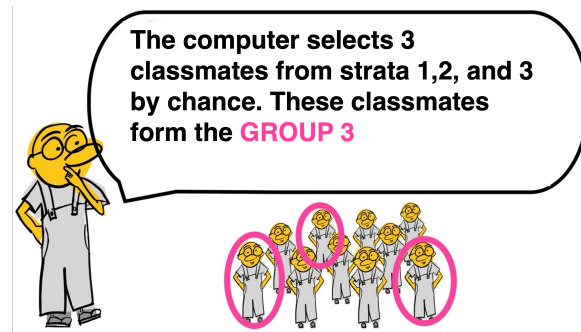
Part 3: Recommendation - Random draw

In this part, the computer will randomly choose three students who belong to strata 1, 2, or 3. We will ask you to nominate three people you think the computer will choose.

We will allocate up to [classroom-specific number equal to 20% of class size] bonuses of 100.000 pesos for Part 3. The steps for allocating the bonuses are explained below.

[classroom-specific illustrations explaining the incentive structure]

Figure A.6: Illustration for the Guessing Task



Select the students in this classroom who belong to strata 1, 2, or 3, who you think will be randomly selected by the computer (Select 3 students).

[Classroom-specific list of all the names visible on one screen. Participants have to pick 3 classmates to continue. Picking their own name invalidates their choices.]

Part 4

Do you want to know your scores on the general intelligence test and the social skills test? We can analyze the data and give you a report that explains your strengths in these two areas. Also, what do these strengths mean, and how can you leverage them for your personal and professional development?

If you want to receive your skills report, we need to contact you again. We also want to be able to invite you to new studies where you can participate for more bonus money. Please indicate if you agree to be contacted again.

[I can be contacted for new studies and to send me my report. I can be contacted to

send my report, but not for new studies. No, I do not want to be contacted again.]

[if participant gives consent to be contacted again]

Please enter your contact email:

[student email]